Image Dehazing using Adaptive Reconstruction

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Introduction

- Images captured in outdoor environments often suffer from degradation due to atmospheric haze.
- Haze is produced when light from scene objects is scattered by particles such as dust, smoke and water droplets.
- This scattering results in
  - reduced contrast
  - loss of scene details
- Image dehazing aims to recover the underlying clean scene from the degraded observation.

Problem Statement

Goal of dehazing:

- Recover the clean scene radiance from a hazy image.
- Estimate two unknown quantities:
  - transmission map
  - atmospheric light

Using the estimated unknown parameters and dehazing the image.

Image Formation under Haze

- This physical process is modeled mathematically using the atmospheric scattering model.
- The haze formation follows the atmospheric scattering model:

$$I(x) = J(x)t(x) + A(1 - t(x))$$

where

- $I(x)$: observed hazy image
- $J(x)$: clean scene radiance
- $t(x)$: transmission map
- A : atmospheric light

Clean Image Recovery

Rearranging the atmospheric model:

$$J(x) = \frac{I(x) - A(1 - t(x))}{t(x)}$$

Problem:

- when transmission $t(x)$ is very small, the division becomes unstable
- leads to noise amplification and artifacts

Solution: introduce regularized reconstruction.

Baseline Reconstruction

To stabilize inversion we use a pseudo-inverse formulation:

$$H_{pinv} = \frac{t}{t^2 + \lambda}$$

Recovered image:

$$J = H_{pinv}(I - A) + A$$

- λ controls reconstruction stability
- prevents amplification when transmission is small

Review 1: Initial Dehazing Framework

In the first stage of the work, a reconstruction-based dehazing approach was implemented.

- Transmission map was estimated from the hazy image.
- Atmospheric light was estimated using the brightest pixels in the image.
- The clean image was reconstructed using a regularized pseudo-inverse formulation.
- A fixed regularization parameter λ was used.
- The gap found was with a fixed λ set, dehazing for thick haze is difficult.

Result Comparison

Input Hazy RGB Image

Final Balanced RGB Output

Update 1: Adaptive Regularization

A fixed regularization parameter cannot handle different haze intensities. Haze strength is estimated as

$$H = \text{mean}(1 - t(x))$$

Based on haze intensity different set of λ values are used:

$$\lambda = [0.05, 0.1, 0.2, 0.35]$$

$$\lambda = [0.2, 0.3, 0.4, 0.6]$$

Each λ produces a reconstruction.

Multi Reconstruction Fusion

Multiple reconstructed images are obtained:

$$J_1, J_2, J_3, J_4$$

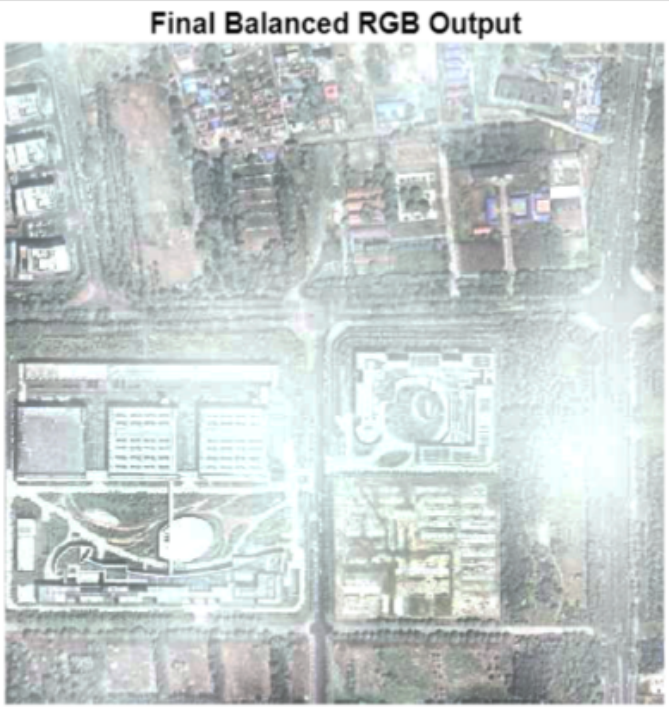
These reconstructions are fused in the frequency domain:

$$F = \mathcal{F}^{-1}\left(\frac{1}{N}\sum_{k=1}^{N}\mathcal{F}(J_k)\right)$$

Advantages:

- combines complementary details
- reduces reconstruction artifacts
- improves image quality

Result Comparison



Update 2: Patch-wise Haze Modeling

Real haze is spatially varying.
To simulate realistic haze:

- select two image regions
- smooth transmission using Gaussian filtering

The hazy image is generated using

$$I = J(x)t(x) + A(1 - t(x))$$

Patch Selective Dehazing

A haze mask M identifies hazy regions.
Final reconstruction:

$$J_{final} = MJ_{corrected} + (1 - M)I$$

- haze regions are corrected
- clear regions remain unchanged
- prevents over-enhancement

Result Comparison

Evaluation

Performance is evaluated using Frobenius norm analysis.

$$||I||_F = \sqrt{\sum I^2}$$

$$||J||_F = \sqrt{\sum J^2}$$

Haze reduction:

$$Reduction = \frac{||I - J||}{||I||} \times 100$$

Conclusion

- Developed a progressive image dehazing framework.
- Introduced adaptive reconstruction and patch haze modeling.
- Evaluated performance using Frobenius norm energy analysis.

Thank You